EXP:02 CHARACTERISTICS STUDY OF SIDE BAND POWER AND FREQUENCIES IN AM USING MATLAB

```
clc;
clear;
close all;

%% ============================================================
% CHARACTERISTICS STUDY OF SIDEBAND POWER AND FREQUENCIES IN AM
% USING MATLAB
% ============================================================

%% 1. INITIAL PARAMETERS

Fs = 1e6;            % Sampling frequency = 1 MHz
T  = 0.01;           % Simulation time = 10 ms

t = 0:1/Fs:T-1/Fs;

% Carrier and message parameters
fc = 100e3;          % Carrier frequency = 100 kHz
fm = 10e3;           % Message frequency = 10 kHz

Ac = 1;              % Carrier amplitude
Am = 1;              % Message amplitude

R = 50;              % Load resistance = 50 ohms

%% 2. MODULATION INDEX

mu = Am/Ac;

fprintf('\n');
fprintf('=====================================================\n');
fprintf('      AM SIDEBAND POWER AND FREQUENCY STUDY\n');
fprintf('=====================================================\n');

fprintf('\nBasic Parameters:\n');
fprintf('Carrier Frequency       = %.2f kHz\n',fc/1000);
fprintf('Message Frequency       = %.2f kHz\n',fm/1000);
fprintf('Carrier Amplitude       = %.2f V\n',Ac);
fprintf('Message Amplitude       = %.2f V\n',Am);
fprintf('Load Resistance         = %.2f Ohm\n',R);
fprintf('Modulation Index        = %.2f\n',mu);

%% 3. MESSAGE SIGNAL

message = Am*cos(2*pi*fm*t);

%% 4. CARRIER SIGNAL

carrier = Ac*cos(2*pi*fc*t);

%% 5. AM SIGNAL

AM_signal = Ac*(1 + mu*cos(2*pi*fm*t)) ...
          .* cos(2*pi*fc*t);

%% ============================================================
% 6. TIME DOMAIN PLOTS
% ============================================================

% Message signal

figure;

plot(t*1000,message,'LineWidth',1.2);

xlabel('Time (ms)');
ylabel('Amplitude (V)');

title('Message Signal');

grid on;

xlim([0 1]);
```

```
% Carrier signal

figure;

plot(t*1000,carrier,'LineWidth',1);

xlabel('Time (ms)');
ylabel('Amplitude (V)');

title('Carrier Signal');

grid on;

xlim([0 0.2]);

% AM signal

figure;

plot(t*1000,AM_signal,'LineWidth',1);

xlabel('Time (ms)');
ylabel('Amplitude (V)');

title('Amplitude Modulated (AM) Signal');

grid on;

xlim([0 1]);

%% ============================================================
% 7. AM SPECTRUM
% ============================================================

N = length(AM_signal);

AM_fft = fft(AM_signal);

AM_fft_shifted = fftshift(AM_fft);

f = (-N/2:N/2-1)*(Fs/N);

AM_magnitude = abs(AM_fft_shifted)/N;

figure;

plot(f/1000,AM_magnitude,'LineWidth',1.2);

xlabel('Frequency (kHz)');
ylabel('Magnitude');

title('AM Frequency Spectrum');

grid on;

xlim([70 130]);

%% ============================================================
% 8. FIND USB AND LSB FREQUENCIES
% ============================================================

f_USB = fc + fm;

f_LSB = fc - fm;

fprintf('\n');
fprintf('Sideband Frequencies:\n');

fprintf('LSB Frequency          = %.2f kHz\n',f_LSB/1000);

fprintf('Carrier Frequency      = %.2f kHz\n',fc/1000);

fprintf('USB Frequency          = %.2f kHz\n',f_USB/1000);
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%% ============================================================
% 9. BANDWIDTH
% ============================================================

BW_AM = 2*fm;

fprintf('\n');
fprintf('AM Bandwidth:\n');

fprintf('BW = 2fm = %.2f kHz\n',BW_AM/1000);

%% ============================================================
% 10. CARRIER POWER
% ============================================================

Pc = Ac^2/(2*R);

fprintf('\n');
fprintf('Carrier Power:\n');

fprintf('Pc = %.6f W\n',Pc);

%% ============================================================
% 11. SIDEband POWER
% ============================================================

P_USB = (mu^2 * Pc)/4;

P_LSB = (mu^2 * Pc)/4;

P_SB_total = P_USB + P_LSB;

%% ============================================================
% 12. TOTAL AM POWER
% ============================================================

P_total = Pc + P_SB_total;

%% ============================================================
% 13. POWER EFFICIENCY
% ============================================================

efficiency = (P_SB_total/P_total)*100;

%% ============================================================
% 14. DISPLAY POWER RESULTS
% ============================================================

fprintf('\n');
fprintf('Power Calculations:\n');

fprintf('Carrier Power           = %.6f W\n',Pc);

fprintf('USB Power               = %.6f W\n',P_USB);

fprintf('LSB Power               = %.6f W\n',P_LSB);

fprintf('Total Sideband Power    = %.6f W\n',P_SB_total);

fprintf('Total AM Power          = %.6f W\n',P_total);

fprintf('Transmission Efficiency = %.2f %%\n',efficiency);

%% ============================================================
% 15. MODULATION INDEX STUDY
% ============================================================

mu_values = 0.1:0.1:1.0;

P_USB_values = zeros(size(mu_values));

P_LSB_values = zeros(size(mu_values));

P_SB_values = zeros(size(mu_values));
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P_total_values = zeros(size(mu_values));

Efficiency_values = zeros(size(mu_values));

for k = 1:length(mu_values)

    m_index = mu_values(k);

    P_USB_values(k) = ...
        (m_index^2 * Pc)/4;

    P_LSB_values(k) = ...
        (m_index^2 * Pc)/4;

    P_SB_values(k) = ...
        P_USB_values(k) + P_LSB_values(k);

    P_total_values(k) = ...
        Pc + P_SB_values(k);

    Efficiency_values(k) = ...
        (P_SB_values(k)/P_total_values(k))*100;

end

%% ============================================================
% 16. DISPLAY MODULATION INDEX TABLE
% ============================================================

fprintf('\n');
fprintf('============================================================\n');
fprintf('      POWER VS MODULATION INDEX\n');
fprintf('============================================================\n');

fprintf('\n');

fprintf(' Mu     USB Power    LSB Power    Total SB     Total AM\n');

fprintf('------------------------------------------------------------\n');

for k = 1:length(mu_values)

    fprintf(' %.1f    %.6f    %.6f    %.6f    %.6f\n', ...
        mu_values(k), ...
        P_USB_values(k), ...
        P_LSB_values(k), ...
        P_SB_values(k), ...
        P_total_values(k));

end

fprintf('------------------------------------------------------------\n');

%% ============================================================
% 17. USB POWER VS MODULATION INDEX
% ============================================================

figure;

plot(mu_values,P_USB_values,'o-','LineWidth',1.5);

xlabel('Modulation Index (\mu)');
ylabel('USB Power (W)');

title('USB Power vs Modulation Index');

grid on;

%% ============================================================
% 18. LSB POWER VS MODULATION INDEX
% ============================================================

figure;

plot(mu_values,P_LSB_values,'s-','LineWidth',1.5);
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```
xlabel('Modulation Index (\mu)');
ylabel('LSB Power (W)');

title('LSB Power vs Modulation Index');

grid on;

%% ============================================================
% 19. TOTAL SIDEBAND POWER VS MODULATION INDEX
% ============================================================

figure;

plot(mu_values,P_SB_values,'^-','LineWidth',1.5);

xlabel('Modulation Index (\mu)');
ylabel('Total Sideband Power (W)');

title('Total Sideband Power vs Modulation Index');

grid on;

%% ============================================================
% 20. TOTAL AM POWER VS MODULATION INDEX
% ============================================================

figure;

plot(mu_values,P_total_values,'o-','LineWidth',1.5);

xlabel('Modulation Index (\mu)');
ylabel('Total AM Power (W)');

title('Total AM Power vs Modulation Index');

grid on;

%% ============================================================
% 21. TRANSMISSION EFFICIENCY
% ============================================================

figure;

plot(mu_values,Efficiency_values,'d-','LineWidth',1.5);

xlabel('Modulation Index (\mu)');
ylabel('Efficiency (%)');

title('AM Transmission Efficiency vs Modulation Index');

grid on;

%% ============================================================
% 22. COMPARE USB AND LSB POWER
% ============================================================

figure;

plot(mu_values,P_USB_values,'o-','LineWidth',1.5);

hold on;

plot(mu_values,P_LSB_values,'s-','LineWidth',1.5);

xlabel('Modulation Index (\mu)');
ylabel('Power (W)');

title('USB and LSB Power Comparison');

legend('USB Power','LSB Power');

grid on;

%% ============================================================
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% 23. AM SPECTRUM FOR DIFFERENT MODULATION INDICES
% ============================================================

mu_plot = [0.25 0.5 0.75 1.0];

figure;

for k = 1:length(mu_plot)

    mu_temp = mu_plot(k);

    AM_temp = Ac*(1 + ...
        mu_temp*cos(2*pi*fm*t)) ...
        .* cos(2*pi*fc*t);

    spectrum_temp = fftshift(fft(AM_temp));

    magnitude_temp = ...
        abs(spectrum_temp)/N;

    subplot(4,1,k);

    plot(f/1000,magnitude_temp);

    xlabel('Frequency (kHz)');
    ylabel('Magnitude');

    title(['AM Spectrum, \mu = ', ...
        num2str(mu_temp)]);

    grid on;

    xlim([70 130]);

end

%% ============================================================
% 24. ENVELOPE OF AM SIGNAL
% ============================================================

envelope_upper = Ac*(1 + mu*cos(2*pi*fm*t));

envelope_lower = -Ac*(1 + mu*cos(2*pi*fm*t));

figure;

plot(t*1000,AM_signal);

hold on;

plot(t*1000,envelope_upper,'--','LineWidth',1.2);

plot(t*1000,envelope_lower,'--','LineWidth',1.2);

xlabel('Time (ms)');
ylabel('Amplitude (V)');

title('AM Signal and Its Envelope');

legend('AM Signal','Upper Envelope','Lower Envelope');

grid on;

xlim([0 1]);

%% ============================================================
% 25. IMPORTANT RESULTS AT 100% MODULATION
% ============================================================

mu_100 = 1;

P_USB_100 = (mu_100^2 * Pc)/4;

P_LSB_100 = (mu_100^2 * Pc)/4;
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```
P_SB_100 = P_USB_100 + P_LSB_100;

P_total_100 = Pc + P_SB_100;

efficiency_100 = ...
    (P_SB_100/P_total_100)*100;

fprintf('\n');
fprintf('=========================================================\n');
fprintf('          RESULTS FOR 100%% MODULATION\n');
fprintf('=========================================================\n');

fprintf('Carrier Power            = %.6f W\n',Pc);

fprintf('USB Power                = %.6f W\n',P_USB_100);

fprintf('LSB Power                = %.6f W\n',P_LSB_100);

fprintf('Total Sideband Power     = %.6f W\n',P_SB_100);

fprintf('Total AM Power           = %.6f W\n',P_total_100);

fprintf('Transmission Efficiency  = %.2f %%\n',efficiency_100);

fprintf('=========================================================\n');

%% ========================================================
% 26. FINAL CONCLUSION
% ========================================================

fprintf('\n');
fprintf('CONCLUSION:\n');

fprintf(['1. USB frequency = fc + fm.\n']);

fprintf(['2. LSB frequency = fc - fm.\n']);

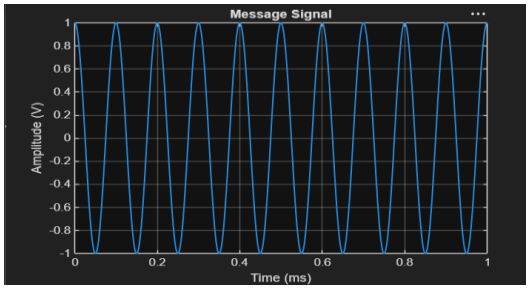
fprintf(['3. AM bandwidth = 2fm.\n']);

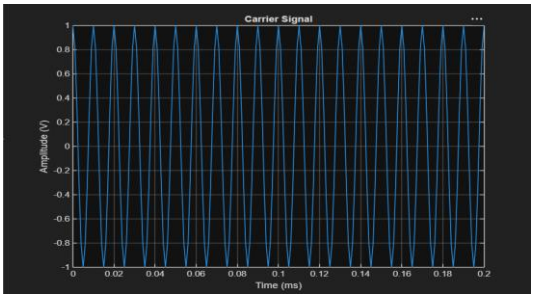
fprintf(['4. Sideband power increases with mu^2.\n']);

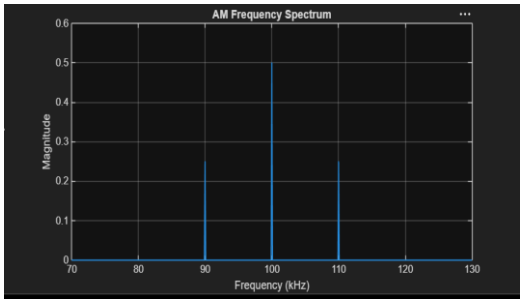
fprintf(['5. At 100%% modulation, total sideband power = Pc/2.\n']);

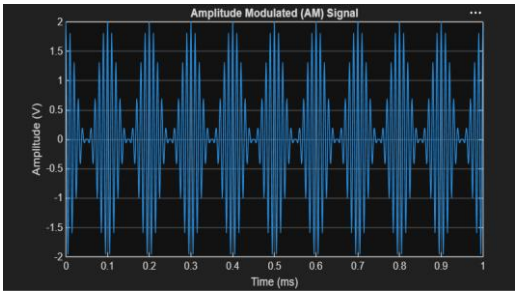
fprintf(['6. At 100%% modulation, total AM power = 1.5Pc.\n']);

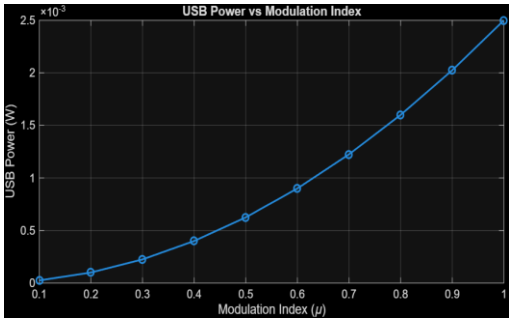
fprintf(['7. Maximum theoretical AM efficiency = 33.33%%.\n']);
fprintf('\n=========================================================\n');
fprintf('              SIMULATION COMPLETED\n');
fprintf('=========================================================\n');
```

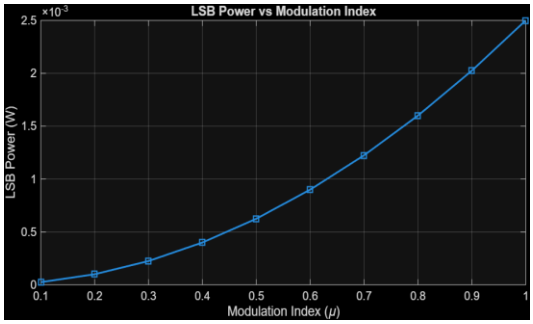


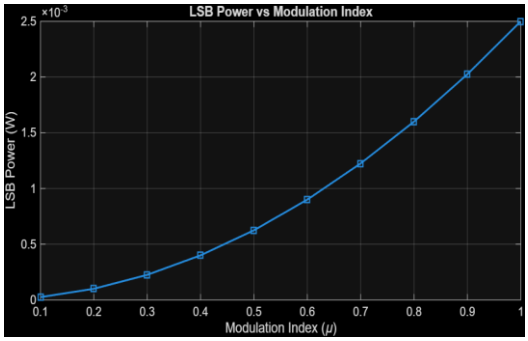


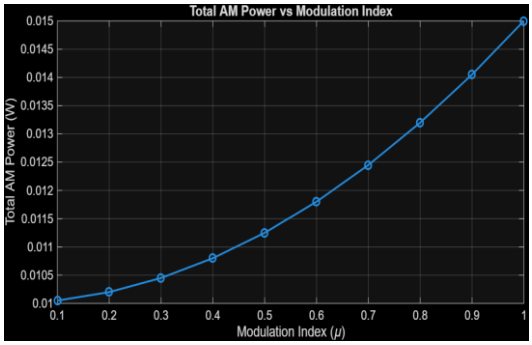


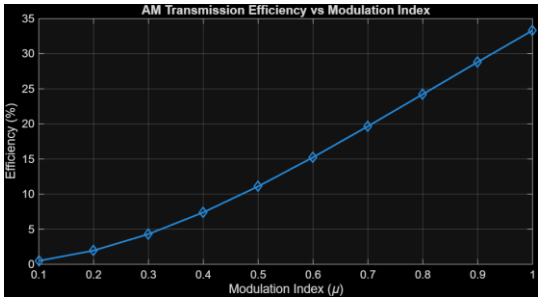


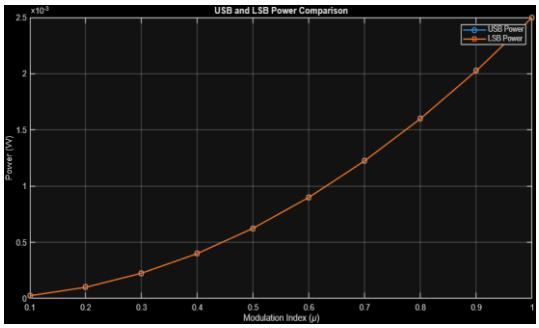

USB Power vs Modulation Index
USB Power (W)
Modulation Index (μ)

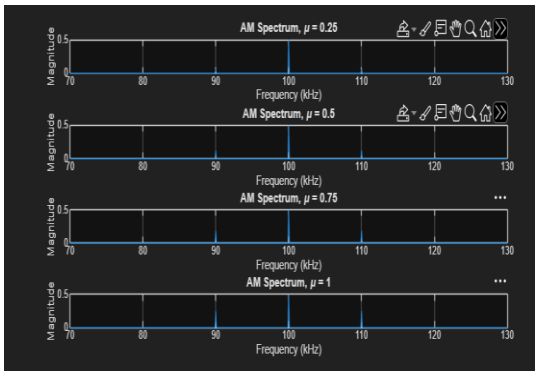

LSB Power vs Modulation Index
LSB Power (W)
Modulation Index (μ)

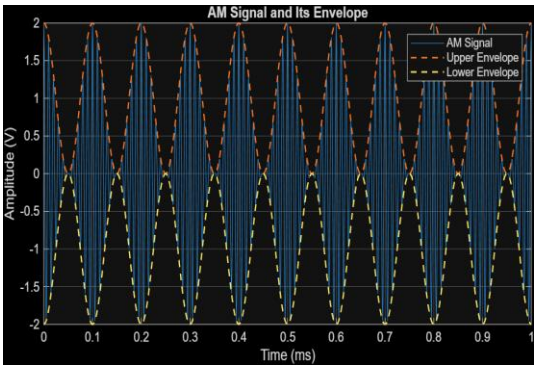

LSB Power vs Modulation Index
LSB Power (W)
Modulation Index (μ)


Total AM Power vs Modulation Index
Total AM Power (W)
Modulation Index (μ)


AM Transmission Efficiency vs Modulation Index
Efficiency (%)
Modulation Index (μ)


USB and LSB Power Comparison
USB Power
LSB Power
Power (W)
Modulation Index (μ)


AM Spectrum, μ = 0.25
AM Spectrum, μ = 0.5
AM Spectrum, μ = 0.75
AM Spectrum, μ = 1
Magnitude
Frequency (kHz)


AM Signal and Its Envelope
AM Signal
Upper Envelope
Lower Envelope
Amplitude (V)
Time (ms)